

Recommendations — osteosarcoma-mets-dll3-h7r2

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** bone
- **Histology:** osteosarcoma
- **Stage:** IV
- **ECOG:** 1
- **Age band:** 18-29
- **Sex:** unknown
- **Biomarkers:** DLL3 RNA expression elevated; protein-level membrane status unknown (rna_only); PRAME RNA expression elevated; protein-level expression and HLA presentation status unknown (rna_only)
- **Efficacy/toxicity weight:** 0.85

4 ranked therapeutic options across 2 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

DLL3-targeting interventions

1. tarlatamab via NCT06788938 (UCCC-01 basket) *(conditional on dll3_ihc positive)*

Likelihood of effect: Cross-tumor extrapolation: SCLC OS HR 0.60 (DeLLphi-304); ORR ~40% (DeLLphi-301). Osteosarcoma efficacy is the open question NCT06788938 will answer.

Toxicity burden: Moderate (CRS ~50% mostly G1-2; CRS G≥3 ~1%; ICANS-like ~10%; inpatient cycle-1 step-up dosing required)

Counter-productive MoA: **Moderate** (On-mechanism CNS bystander T-cell activation drives ICANS; possible DLL3 antigen-loss escape on repeated dosing)

Overall: The only DLL3-directed option when IHC is positive — preference-aligned but cross-tumor translation untested; foreclosed if IHC negative.

Key references: NCT06788938 · PMID 37861218 · PMID 40454646

2. SHR-4849 / IDE849 via NCT07174583 (IDEAYA pan-tumor DLL3 basket) *(conditional on dll3_ihc positive)*

Likelihood of effect: Speculative. No published clinical data for SHR-4849. TOP1-inhibitor-payload ADC precedent (T-DXd, sacituzumab govitecan) suggests the payload class can produce response signals, but DLL3-specific efficacy is the open question.

Toxicity burden: Unknown — no published clinical data. TOP1-inhibitor ADC class effects expected: cytopenias, GI toxicity, possible ILD/pneumonitis (DXd-class ADCs carry an ILD signal worth monitoring).

Counter-productive MoA: Moderate (ADC bystander toxicity to DLL3-low normal tissue; antigen-loss escape on repeated dosing; PBD-payload class shadow (Rova-T TAHOE) does not directly apply but informs the toxicity-budget framing.)

Overall: Mechanism-distinct second DLL3 pathway pending sponsor confirmation of osteosarcoma eligibility — relevant if the tarlatamab path is foreclosed or fails.

Key references: NCT07174583

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Modality	Phase	Enrolling indication	Recruitment	Trial	
trispecific	3	relapsed SCLC	recruiting	NCT07189455	
ADC	3	small-cell lung cancer	recruiting	NCT07218146	
BiTE	3	extensive-stage SCLC (1L)	recruiting	NCT07472517	
trispecific	1/2	DLL3-expressing SCLC, large-cell NEC, neuroendocrine prostate cancer, GEP-NEC	active not recruiting	NCT04471727	
bispecific_other	1/2	SCLC, LCNEC, NEPC, GEP-NEC, EP-NEC	recruiting	NCT05652686	
BiTE	1	advanced solid tumors with DLL3 expression	unknown	NCT05461287	
cells	CAR-NK	1	extensive-stage SCLC	unknown	NCT05507593
	trispecific	1	SCLC, neuroendocrine carcinoma	completed	NCT05619744
CAR-T	1	extensive-stage SCLC, LCNEC of lung	active not recruiting	NCT05680922	
ADC	1	advanced solid tumors, SCLC, LCNEC	recruiting	NCT06424665	
ADC	1	small-cell lung cancer	recruiting	NCT06613009	

	radioligand	1	SCLC, large-cell neuroendocrine carcinoma	active not recruiting	NCT06736418
	radioimmunotherapy	1	neuroendocrine tumors / carcinomas (lung, prostate)	not yet recruiting	NCT06941480
	radioligand	1	SCLC, LCNEC, NEPC, GEP-NEC	recruiting	NCT07006727
	ADC	3	small-cell lung cancer (2L)	completed	NCT03061812
	ADC	1	small-cell lung cancer	terminated	NCT02500914
AMG 119	CAR-T	1	small-cell lung cancer	unknown	NCT03392064

Evidence in detail

One row per published clinical manuscript supporting each ranked intervention, drawn from the case's clinical-evidence dossier.

tarlatamab

n	Toxicities	Efficacy	Reference
220	CRS (grade any): 51%; CRS (grade >=3): 1%; ICANS-like neurologic event (grade any): 10%; treatment-related AE (grade >=3): 30%; treatment discontinuation due to AE (grade any): 3%	ORR 40% (95% CI 29-52); DoR not reached months; PFS 4.9 months; TRAE_rate 30%	Ahn et al. N Engl J Med 2023
SCLC (2L+)			
—			
SCLC after			
and			
≥1			
prior			
line;			
ECOG			
0-1;			
brain			
mets			
if			

SCLC 509 after (2L) — or SCLC; ECOG 0-1	CRS (grade any): 56% (n=254); CRS (grade ≥3): 1% (n=254); ICANS-like neurologic event (grade any): 12% (n=254); treatment-related AE (grade ≥3): 24% (n=254); neutropenia (grade ≥3): 3% (n=254)	HR_OS 0.6 HR (95% CI 0.47-0.77); OS 13.6 months; OS 8.3 months; PFS 4.2 months; PFS 3.2 months; TRAE_rate 24%	Mountzios et al. N Engl J Med 2025
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PRAME-targeting interventions

1. IMA203 (PRAME-TCR-T) via NCT03686124 (Immatics ACTengine pan-solid basket) *(conditional on prame_ihc_hla positive)*

Likelihood of effect: Strong. ORR 54% (95% CI 34-73) in pan-solid PRAME+ HLA-A*02:01+ cohort (Wermke 2024 Lancet Oncol, n=28). Sarcoma cohorts on protocol; osteosarcoma fit unconfirmed but eligibility is mechanism-driven, not tumor-restricted.

Toxicity burden: High (CRS ~100% — G3-4 ~10%; ICANS-like ~25%; uniform post-Cy/Flu cytopenias; one treatment-related death in published cohort; CAR-T-style infusion infrastructure required)

Counter-productive MoA:**Moderate** (On-target / off-tumor toxicity to PRAME-expressing normal testis is class-managed (testis is immune-privileged); CRS / neurotoxicity from T-cell activation is the main mechanism-level risk)

Overall:**Best-evidenced PRAME pathway with pan-solid sarcoma-inclusive basket; conditional on PRAME IHC + HLA-A*02:01 typing both confirming.**

Key references: NCT03686124 · PMID 38821093

2. IMC-P115C via NCT07156136 (Immunocore next-gen PRAME ImmTAC pan-tumor) *(conditional on prame_ihc_hla positive)*

Likelihood of effect: Speculative. No published clinical data for IMC-P115C; relies on brenetafusp class precedent (ORR ~9% in heavily pretreated melanoma; durable in subset).

Toxicity burden: Moderate (ImmTAC class: CRS ~85% mostly G1-2; rash ~70%; transient hypotension; pre-medication-managed)

Counter-productive MoA:**Moderate** (ImmTAC on-target / off-tumor signal in PRAME-expressing normal tissue (low; testis-restricted); CRS from T-cell activation)

Overall:**Mechanism-class alternative to IMA203 in the PRAME space, contingent on PRAME + HLA confirmation and sponsor confirmation osteosarcoma is in scope.**

Key references: NCT07156136 · PMID 39007852

Pipeline context — not currently enrollable

Other agents in this target class drawn from the case dossier. The patient cannot currently enroll in these trials, but they are included for context on the broader modality landscape.

Modality	Phase	Enrolling indication	Recruitment	Trial
bispecific_other	1/2	advanced solid tumors expressing PRAME (cutaneous and uveal melanoma, ovarian, lung, sarcoma cohorts)	active not recruiting	NCT04262466
bispecific_other	1/2	refractory / recurrent solid tumors expressing PRAME	recruiting	NCT05958121
bispecific_other	3	previously untreated advanced cutaneous melanoma	recruiting	NCT06112314
CAR-T	1/2	AML, MDS, uveal melanoma	terminated	NCT02743611

Evidence in detail

One row per published clinical manuscript supporting each ranked intervention, drawn from the case's clinical-evidence dossier.

IMA203 (PRAME-TCR-T)

n	Toxicities	Efficacy	Reference
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28	CRS (grade any): 100% (n=28); CRS (grade >=3): 11% (n=28); ICANS-like neurotoxicity (grade any): 25% (n=28); post-Cy/Flu cytopenia (grade >=3): 100% (n=28); treatment-related death (grade 5): 4% (n=28)	ORR 54% (95% CI 34-73); Wermke et al. Lancet Oncol 2024 DoR not reached months; PFS 5.1 months; TRAE_rate
solid		
head and neck,		
(2L+) —		
subset of		
cohort		
Cy/Flu prior to		

IMC-P115C

n	Toxicities	Efficacy	Reference
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132	CRS (grade any): 85% (n=132); CRS (grade >=3): 3% (n=132); rash (grade any): 70% (n=132); transient hypotension (grade any): 35% (n=132); any treatment-related AE (grade >=3): 30% (n=132)	ORR 9% (95% CI 4-17); PFS 2.1 months; DoR >6 months; TRAE_rate 30%	Hamid et al. Cancer Discov 2024
(2L+)			
—			
subset with prior			
by IHC			
